# Jiasheng Guo

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## EDUCATION

**University of California, Santa Barbara** Expected June 2029

*B.S. Computer Science and Applied Mathematics*

- Research Interests: LLM agents and agent harnesses, RL post-training, world models, generative models
- Coursework: Probability & Statistics, Applied Stochastic Processes, Data Structures & Algorithms, Numerical Algorithms, Combinatorial Analysis, Linear Algebra

## PUBLICATIONS

- Kenan Tang, **Jiasheng Guo**, Jeffrey Lin, Yao Qin, *ExpressEdit: Fast Editing of Stylized Facial Expressions with Diffusion Models in Photoshop*, CVPR 2026 Workshop on Generative AI for Storytelling (AISTORY)
- Minqi Cheng*, **Jiasheng Guo***, *Analysis of the Principle and Two Applications for Monte-Carlo Simulations*, Highlights in Science, Engineering and Technology (IFMPT 2024)

## RESEARCH EXPERIENCE

**Qin Lab, UC Santa Barbara** July 2026 – Present

*Undergraduate Researcher, LLM Agents & World Models* Advisor: Prof. Yao Qin

- Building a proactive smart-glasses assistant that adapts to its wearer, including users with memory or vision impairments; post-trained with RL and evaluated in closed-loop simulation
- Wrote a suite of long-horizon GUI tasks for testing multimodal agents under one shared harness, with Qwen baselines
- Turned the 13 state variables and coupled ODEs of the UVA/Padova type 1 diabetes simulator into a causal map of glucose–insulin dynamics that an LLM can reason over

**Qin Lab, UC Santa Barbara** September 2025 – August 2026

*Undergraduate Researcher, Diffusion-Based Image Editing* Advisor: Prof. Yao Qin

- Released the 135-tag expression dataset for ExpressEdit, built its RAG demo on Hugging Face, and measured editing latency at under 3 s per edit on one consumer GPU
- Compared DistriFusion's spatial parallelism for U-Nets with PipeFusion's pipeline parallelism for DiTs in multi-GPU diffusion inference, on communication overhead and stale features
- Defined JSON-based ImagenWorld metrics for prompt adherence, physical plausibility, and spatial accuracy
- Studied symmetry and alignment biases in iterative image editing, and proposed adaptations of Dynamic Routing and LayerComposer that reduce attention over-smoothing and feature leakage

**UC Berkeley** May 2023 – September 2023

*Student Researcher, Stochastic Processes & Monte Carlo Simulation* Advisor: Prof. F. Alberto Grünbaum

- Studied convergence and variance reduction for Monte Carlo estimators; co-first-authored a paper at IFMPT 2024
- Priced derivatives under Black–Scholes–Merton with stochastic differential equations, and inferred social network structure using MCMC on exponential random graph models

## AWARDS

- Euclid Mathematics Contest Global Honor Roll (Top 80 of 27,650) 2024
- USA Computing Olympiad (USACO) Gold Division 2024
- AMC 12 AIME Qualifier 2024
- Physics Bowl Honorable Mention (Division 2) 2024

## SKILLS & LANGUAGES

**Programming Languages:** Python, C++, R, JavaScript
**Tools:** PyTorch, Hugging Face, ComfyUI, Jupyter
**Languages:** English, Mandarin